

**Erasmus Mundus Joint Master Artificial Intelligence for  
Sustainable Societies (AISS)**

# **Topics on Machine Learning and its Applications**

## **Introduction to knowledge discovery**

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# Introduction to knowledge discovery

Knowledge discovery, data mining and machine learning

Areas of machine learning

The KDD process

The CRISP-DM process

Software development processes: Waterfall and Scrum



# Knowledge Discovery in Databases (KDD)

“

KDD is the non-trivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data (*Fayyad et al., 1996*)

“

Extracting implicit, previously unknown, and likely valuable information from data, typically devising models and patterns from large databases (*Sarker et al., 2000; Bramer, 2007*)

”

”

KDD focuses on extracting **valuable** and **hidden** insights from data, often involving model and pattern creation from large databases.



# Data mining (DM)

“

Merges conventional data analysis methods with algorithms that, within acceptable computational efficiency constraints, identify patterns over vast amounts of data (*Tan et al., 2006*)

”

DM uses **algorithms** to efficiently identify patterns in vast amounts of data, often combining traditional data analysis methods

While often used interchangeably with KDD, DM is a more focused process within the broader KDD process





# Machine Learning (ML)

“

Enables computers to learn from data without explicit programming, uncovering knowledge that would be challenging or impossible to deduce through traditional statistical analysis or manual human data inspection (*Mitchell, 1997*)

”

“

The process in which computer systems use computer science and statistical techniques to progressively improve their performance on data analytical tasks by learning from data (*Samuel, 1959*)

”

While often considered a subfield of DM, the differentiator lies in its emphasis on **automation** as traditional data mining does not necessarily require this automated learning aspect



# DM and ML as intertwined concepts

In the literature, the terms *data mining* and *machine learning* are often used interchangeably.

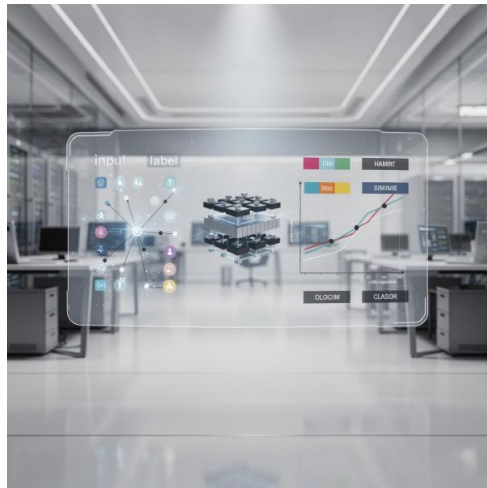
Due to classical statistics having been the cornerstone of both DM and ML, it is not entirely feasible to distinctly separate these two domains.

“They have, in fact, mutually influenced, inspired, learned from, and adopted each other throughout their development and evolution (*Shu and Ye, 2023*)”

“When the term was introduced in the late 1950s, ML represented a transformative shift in data analysis, offering the ability to construct models from algorithms that iteratively learn and refine themselves based on data (*Shu and Ye, 2023*).”

# Areas of Machine Learning

ML is commonly divided into two major areas each encompassing a wide range of algorithms and techniques.



## Supervised learning

Make predictions about new data related to a specific phenomenon



## Unsupervised learning

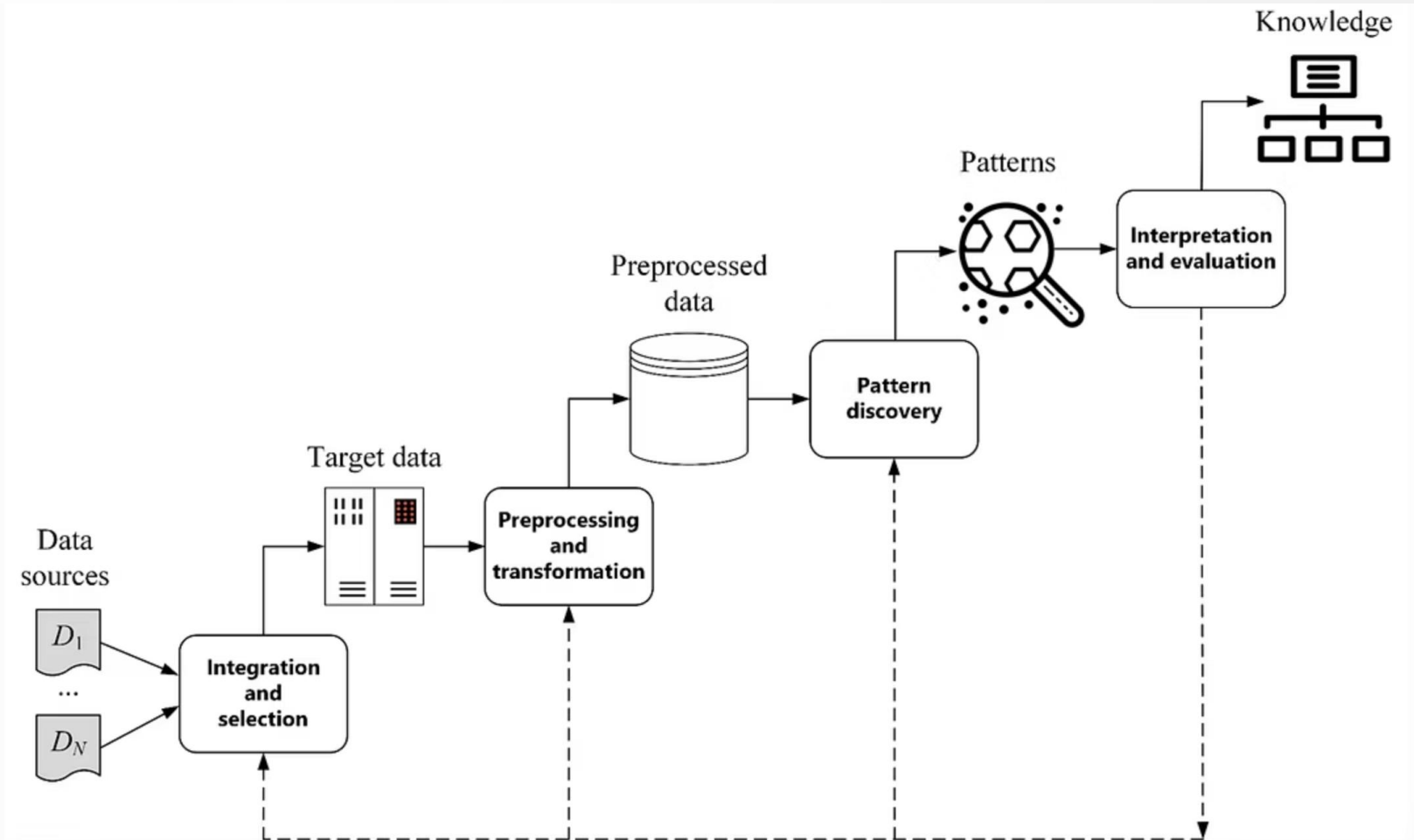
Reveal relationships and patterns within the data that otherwise might not be immediately apparent

# The KDD process

Usama Fayyad, Gregory Piatetsky-Shapiro, and Padhraic Smyth.

*Knowledge Discovery and Data Mining: Towards a Unifying Framework*. In 2nd International Conference on Knowledge Discovery and Data Mining, 1996

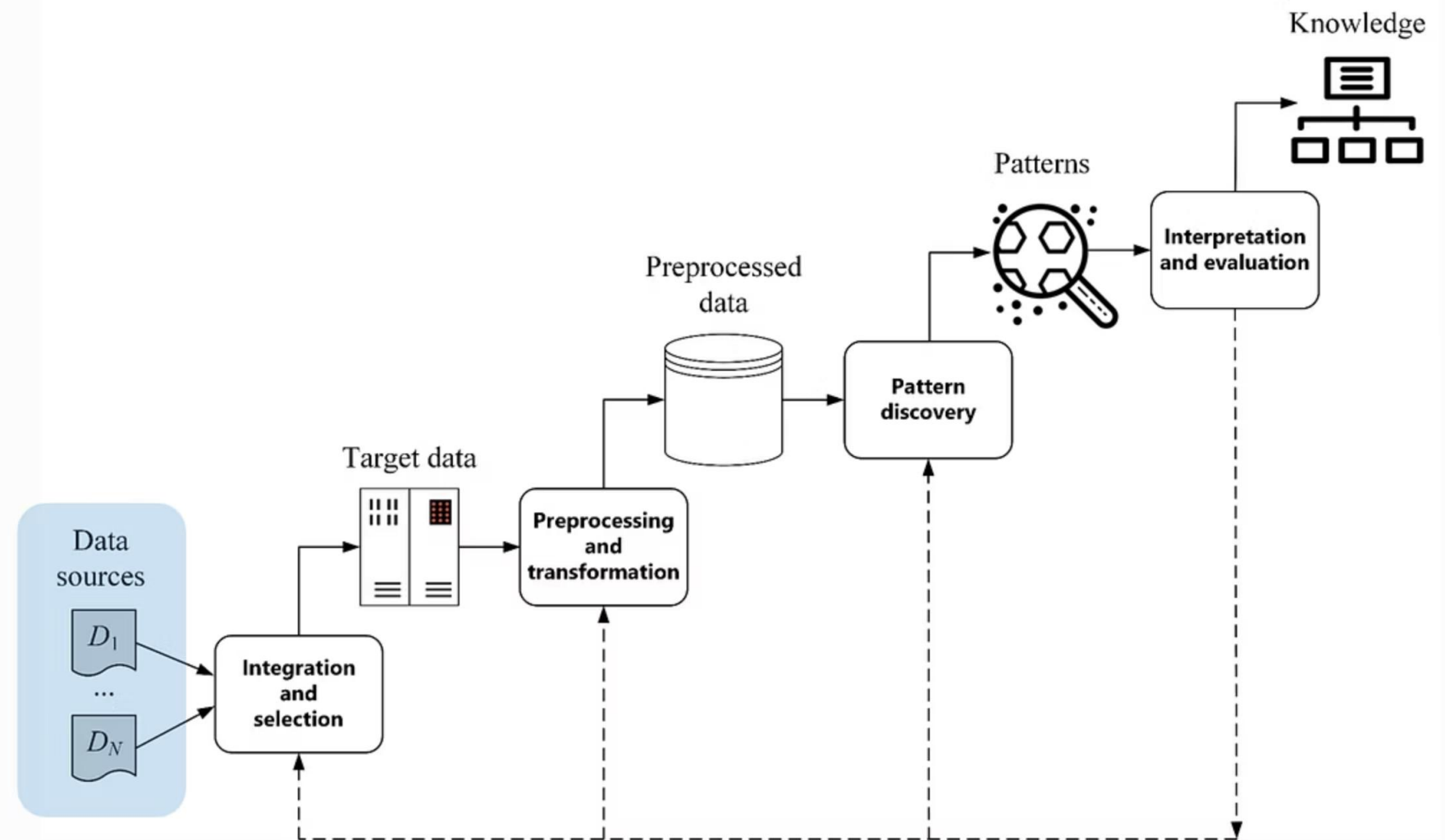
<https://cdn.aaai.org/KDD/1996/KDD96-014.pdf>





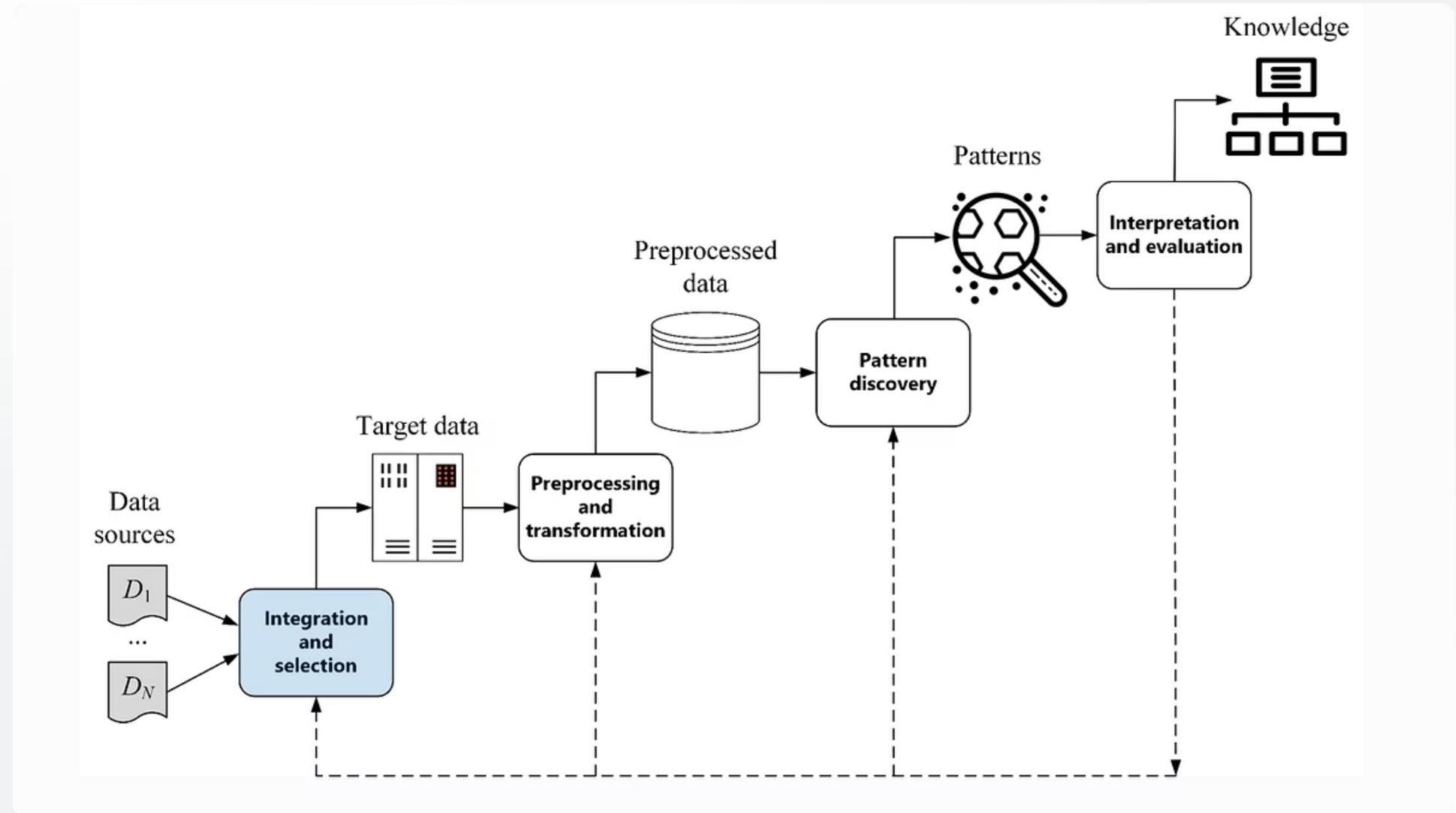
# The KDD process

- The process begins with **data collection** from a variety of sources, diverse formats such as structured databases, unstructured text documents, multimedia files, and sensor data
- The **extraction** can be performed either manually or automatically using a range of techniques



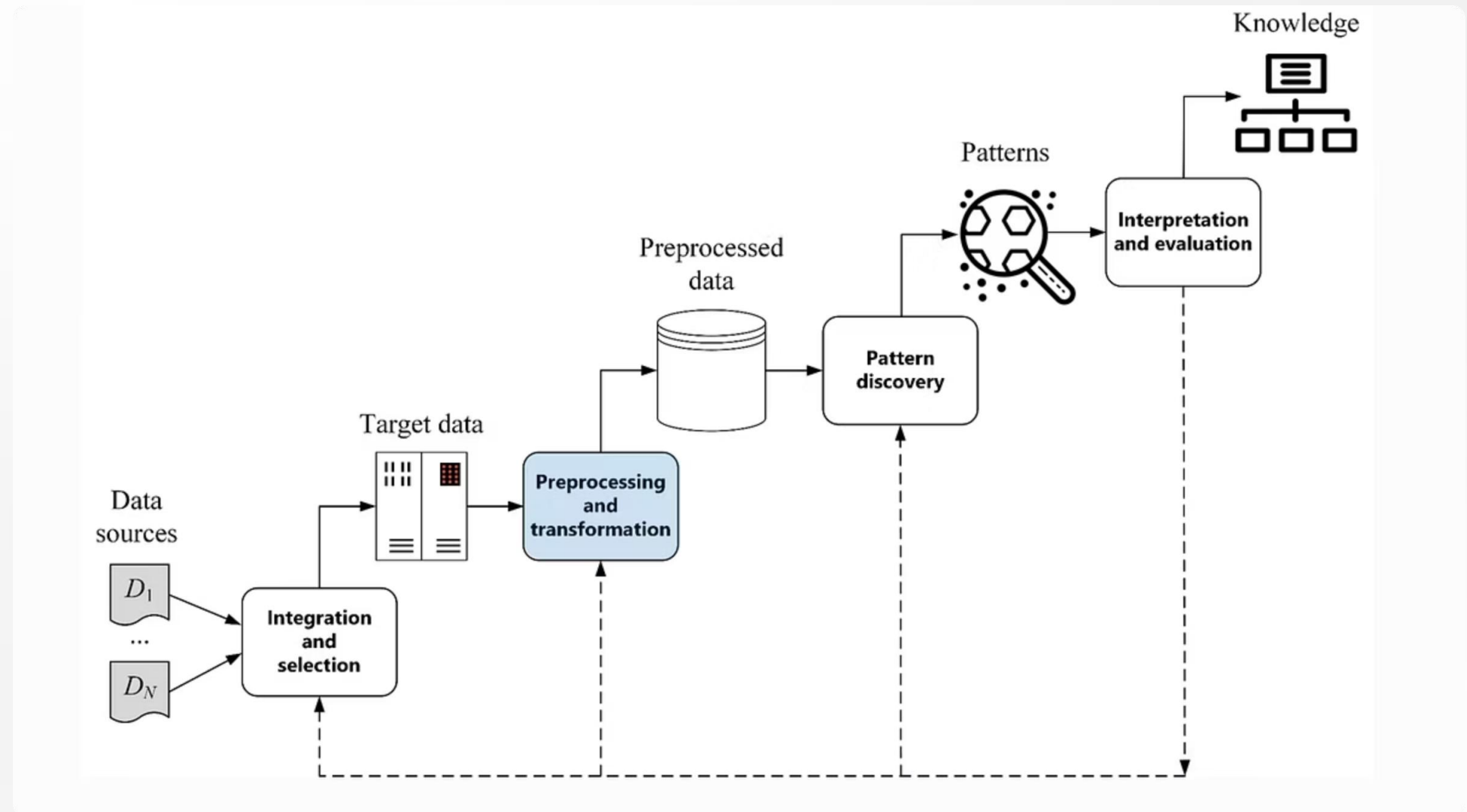
# The KDD process

- Due to the inherent heterogeneity of data sources, an initial stage of **data integration and selection** is often necessary
- This stage involves consolidating data from disparate locations, databases, and systems into a designated repository



# The KDD process

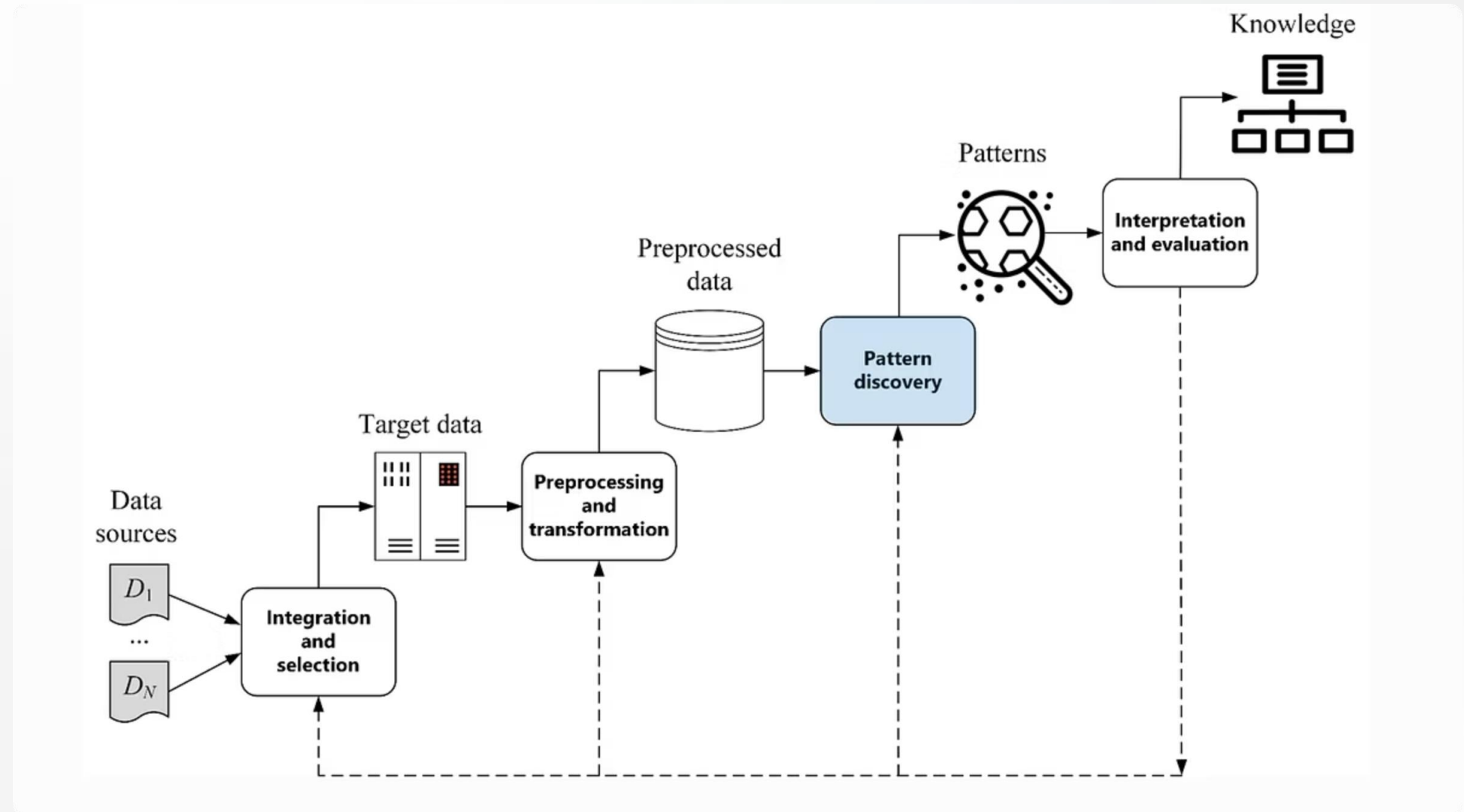
- The chosen dataset undergoes **preprocessing**, an important stage designed to improve its suitability for automated analysis
- Preprocessing encompasses various tasks, including **noise removal**, **handling missing data**, and defining strategies for managing incomplete information





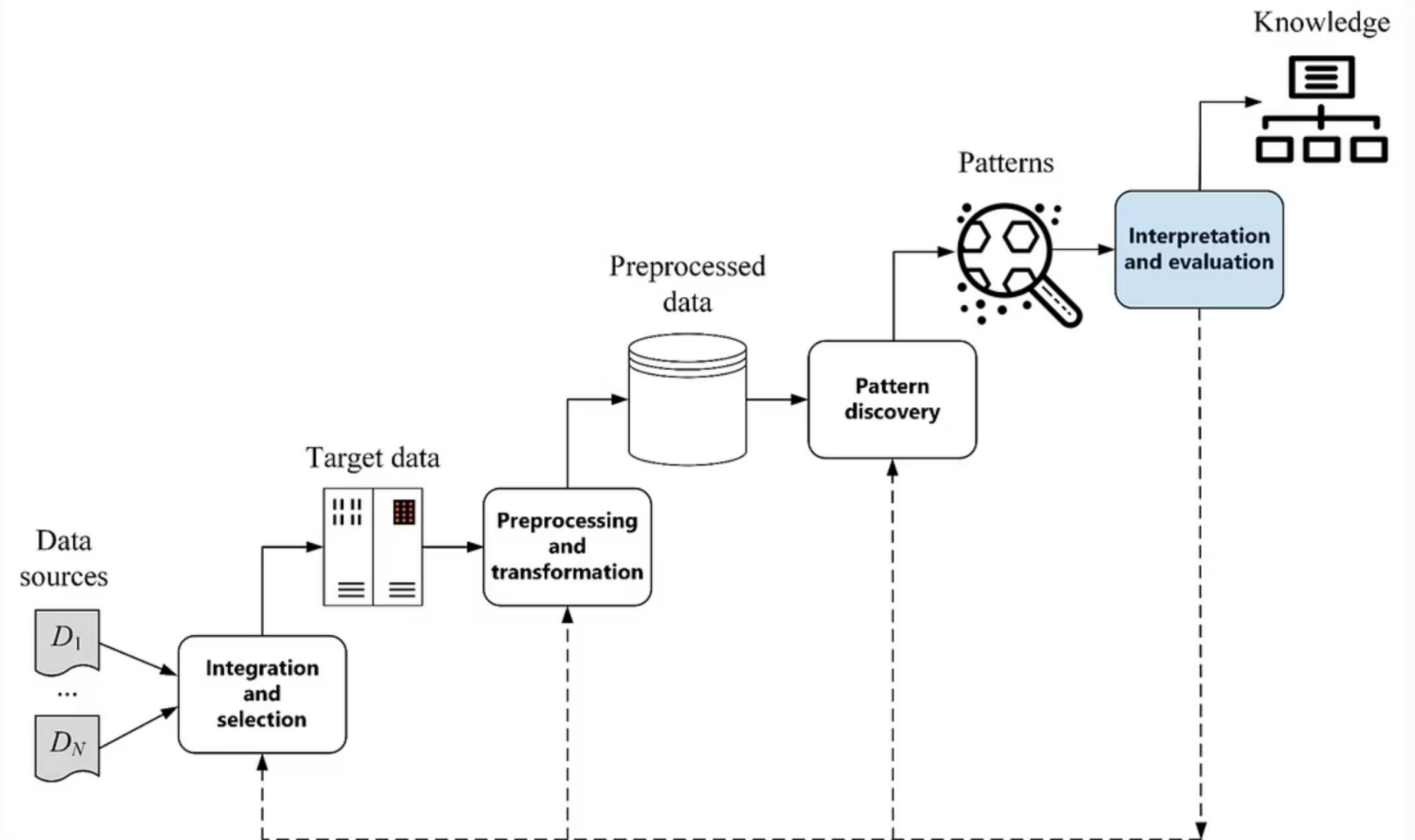
# The KDD process

- **Pattern discovery** aims to uncover latent patterns and relationships within the data, using ML algorithms or various data mining techniques, including manual analysis when appropriate
- Hidden structure and relationships within the data are represented as models or rules



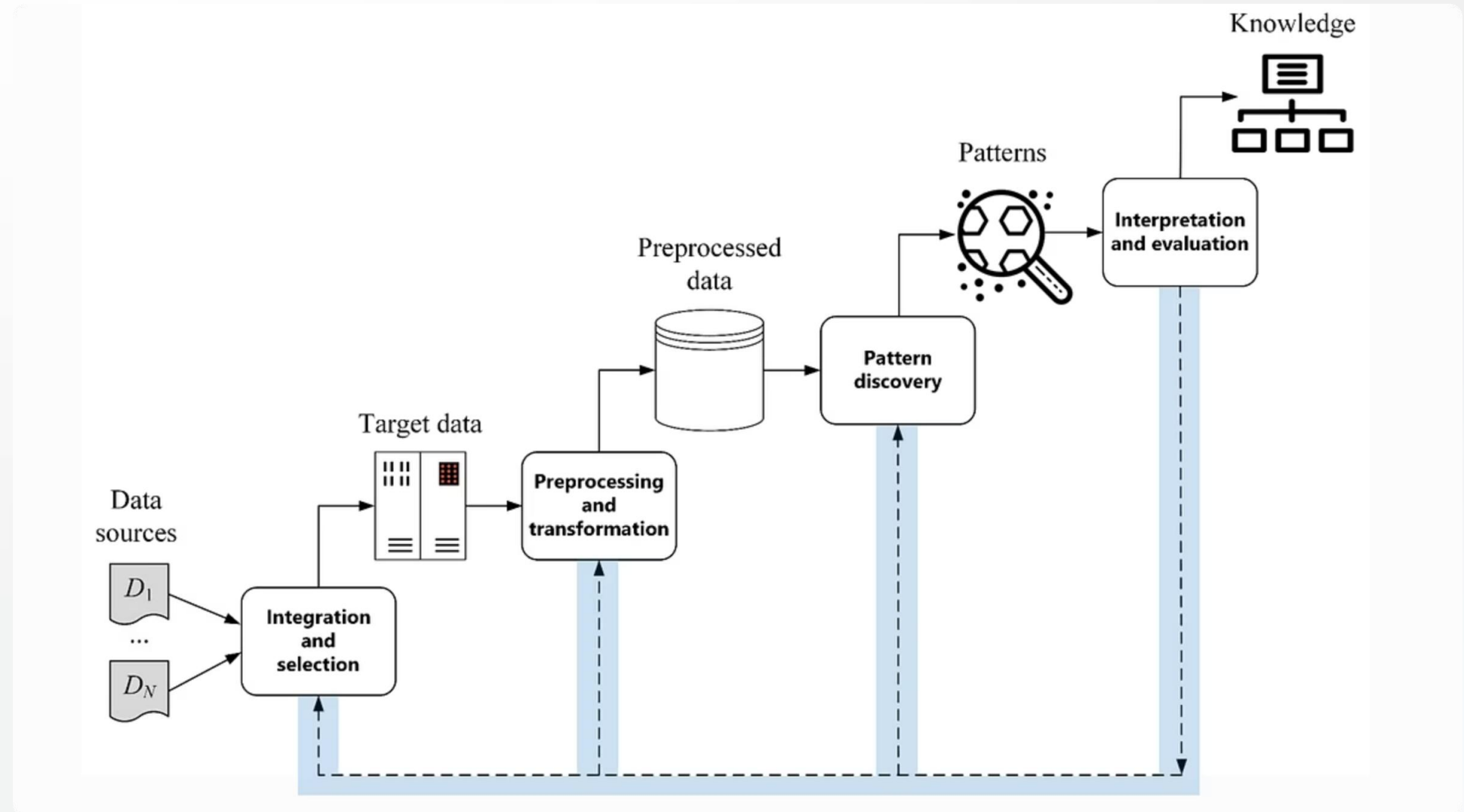
# The KDD process

- Interpretation and evaluation assess the validity and significance of the extracted knowledge
- Evaluation techniques are employed to ensure the patterns accurately reflect the underlying data and align with the research objectives



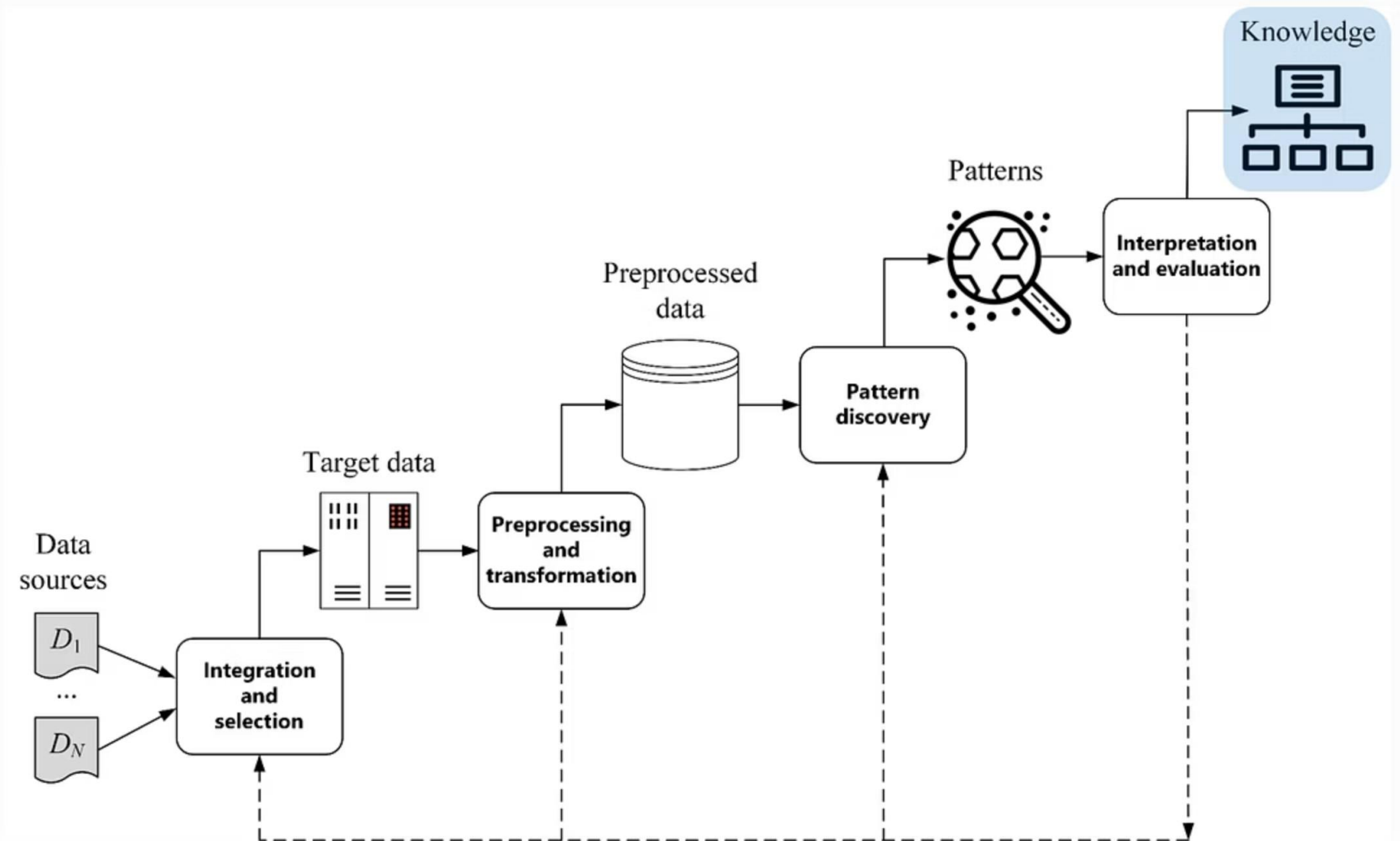
# The KDD process

- If necessary, this stage may necessitate **revisiting previous stages** for further data preparation, algorithm selection, or refinement of the analysis techniques
- This iterative approach ensures the extracted knowledge is not only meaningful but also reliable and generalizable



# The KDD process

- Finally, the KDD process culminates in the consolidation of **discovered knowledge**
- This involves the integration of knowledge into other systems for automated decision-making, visualizing the patterns for enhanced comprehension, or documenting it for future reference



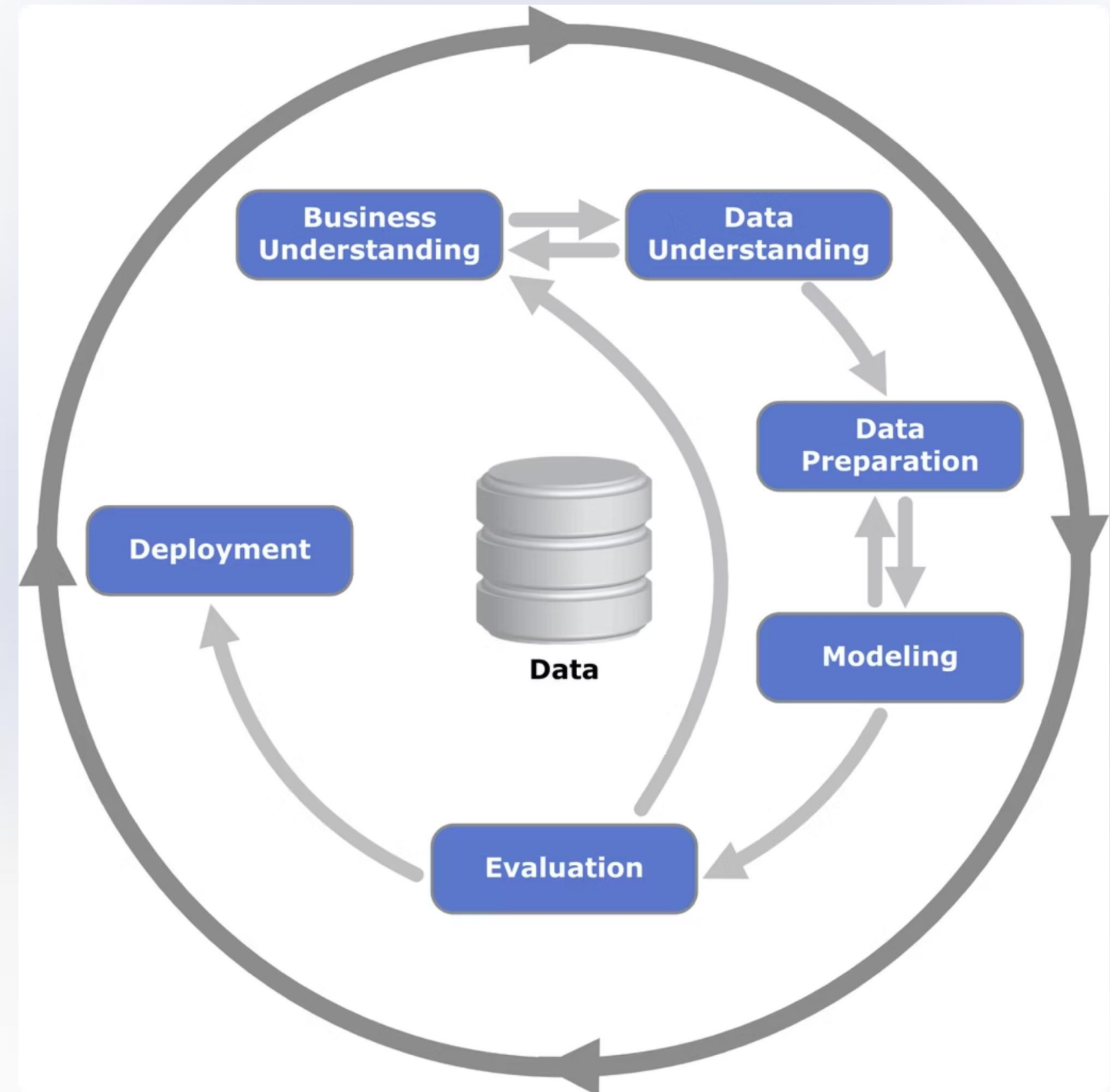


# The CRISP-DM process

Cross-Industry Standard Process for Data Mining  
(CRISP-DM )

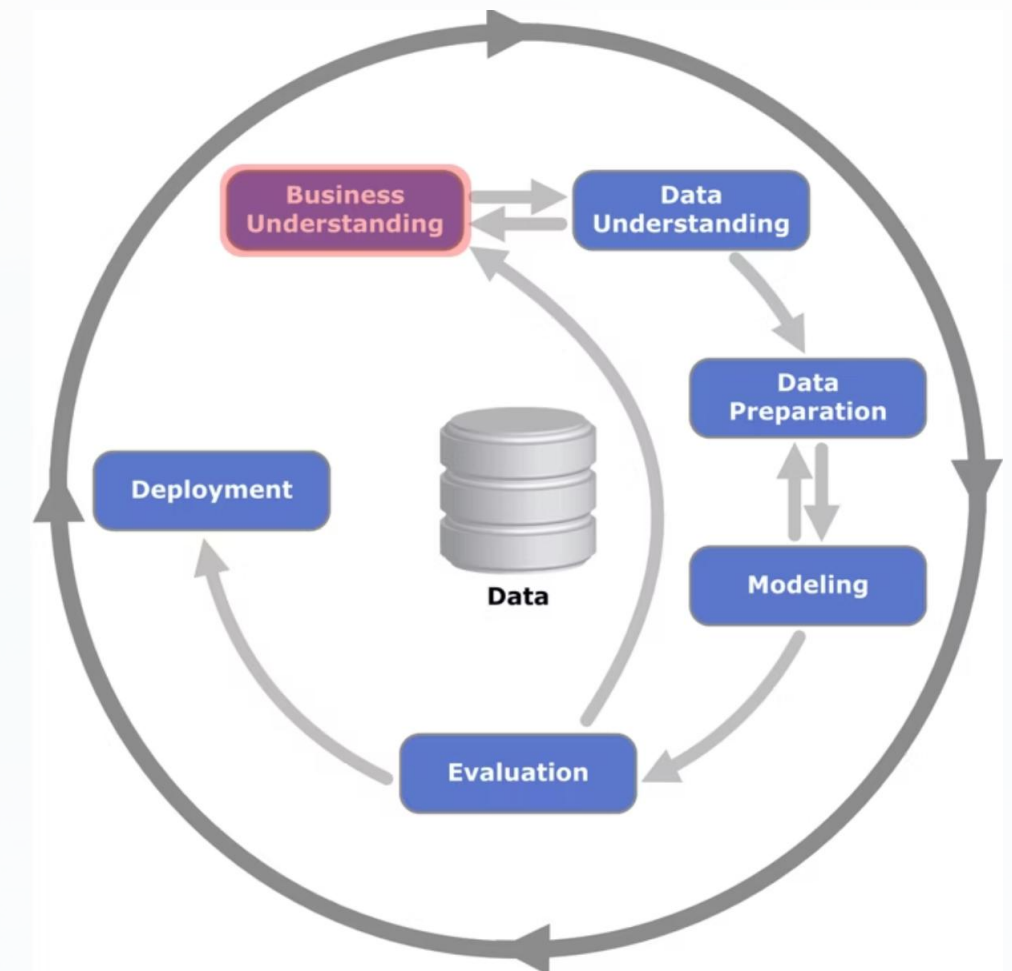
Pete Chapman (NCR Systems Engineering  
Copenhagen). *The CRISP-DM User Guide*. 4th CRISP-  
DM SIG Workshop in Brussels in March 1999

<https://s2.smu.edu/~mhd/8331f03/crisp.pdf>



# The CRISP-DM process

- **Business understanding** focuses on bridging the gap between **business needs** and **data-driven solutions**
- It starts with understanding the project's objectives and requirements from a **business perspective**
- This understanding is then translated into a clearly defined **data mining problem** and a preliminary plan to achieve the desired outcomes
- A formal **decision model**, such as one created using the Decision Model and Notation (DMN) standard, can be a valuable tool in this process to ensure clarity and alignment between business goals and data mining activities



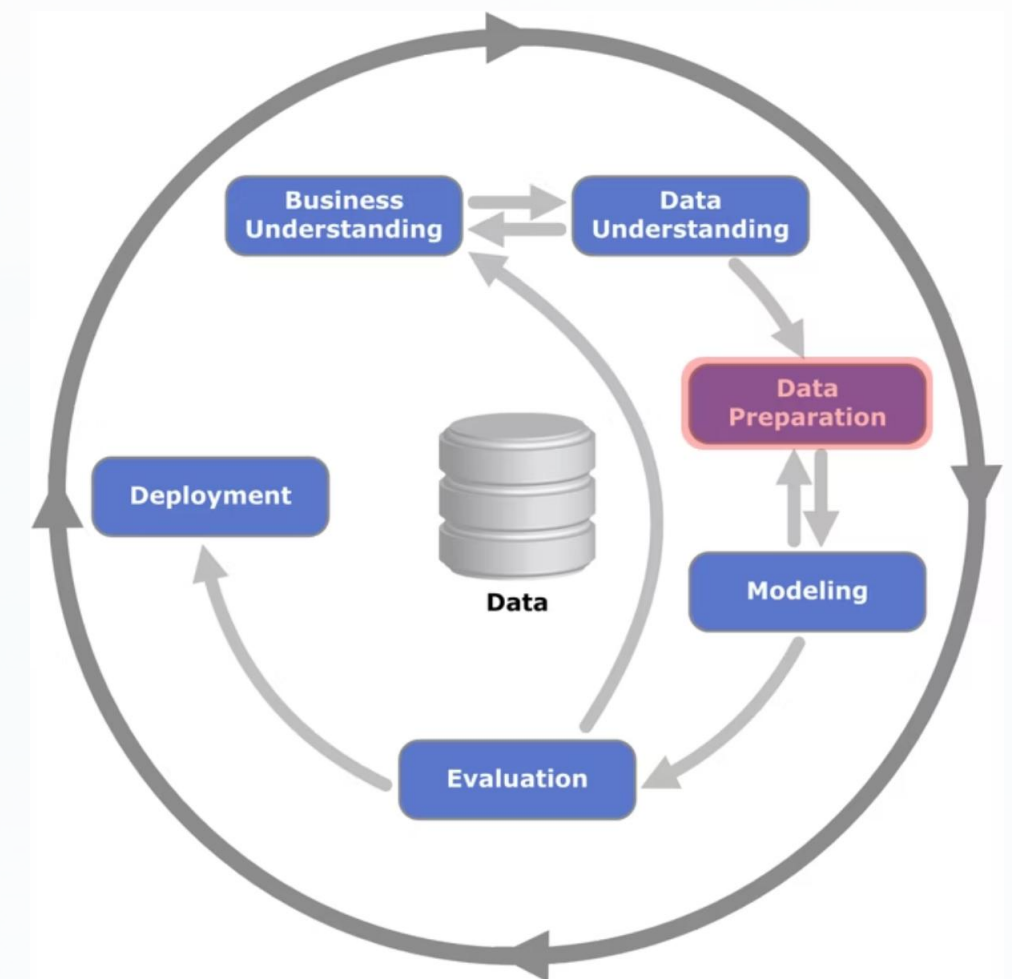
# The CRISP-DM process

- The **data understanding** phase begins with initial data collection and then dives into exploring and assessing that data
- This involves **getting familiar with its structure**, identifying potential quality issues, uncovering preliminary insights, and pinpointing interesting subsets to form hypotheses for hidden information
- This phase allows analysts to **grasp the data's characteristics** and validate its reliability as a foundation for drawing valid inferences and building robust models
- A significant portion of time should be dedicated to this phase, as a deep understanding of the data is essential for mitigating risks and making informed business decisions based on subsequent analysis



# The CRISP-DM process

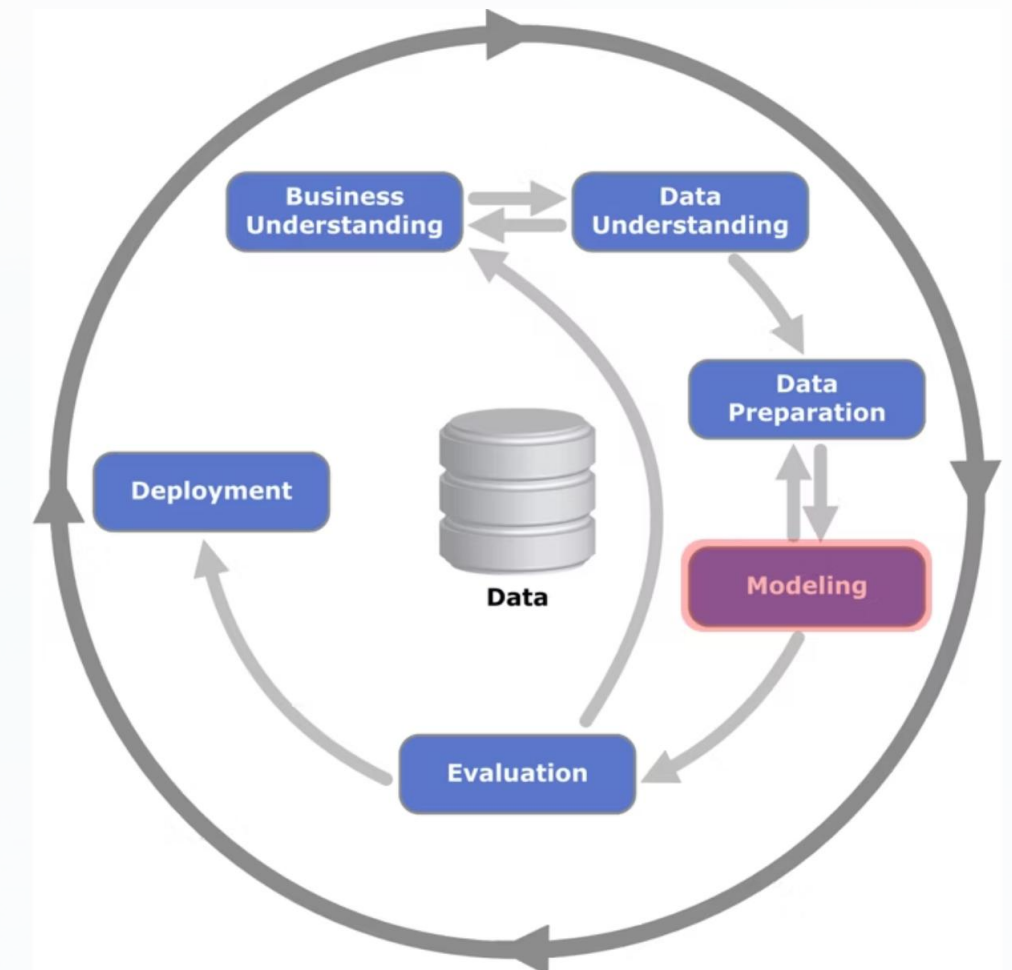
- The **data preparation** phase transforms raw data into a refined dataset ready for analysis. This involves a series of steps tailored to the specific data, such as:
  1. **Consolidating** data from various sources;
  2. **Standardizing** column names for consistency;
  3. **Cleaning** data by addressing errors and inconsistencies;
  4. **Identifying and handling** outliers;
  5. **Transforming** data into a suitable format for analysis tools;
  6. **Partitioning** data into training, testing, and validation sets.
- Once finalized, these cleaning and transformation steps are often automated within an **ETL** (Extract, Transform, Load) process for efficiency and reproducibility





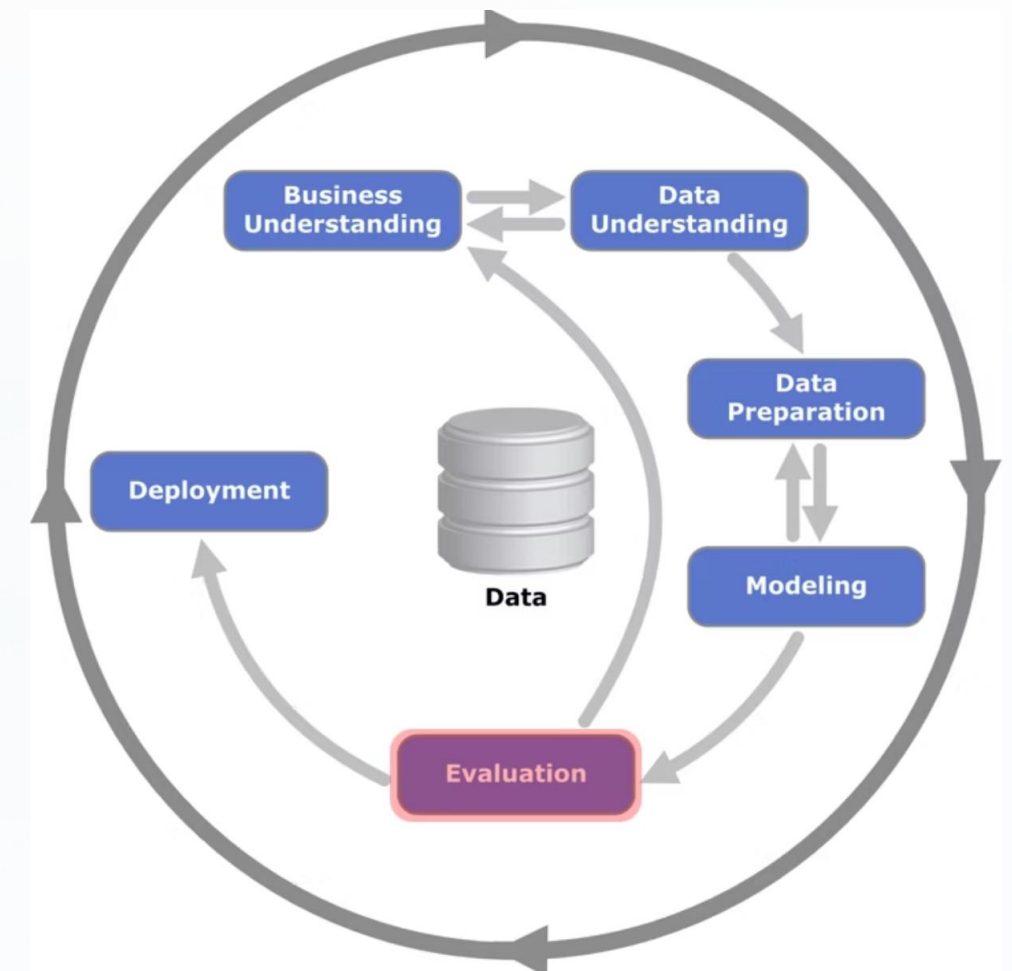
# The CRISP-DM process

- The **modeling phase** involves selecting, applying, and optimizing various modeling techniques to the prepared dataset
- Since different algorithms excel in different scenarios, **multiple techniques are explored** to identify the most effective approach for the specific data mining problem
- Each technique may have unique requirements, sometimes necessitating revisiting the data preparation phase for adjustments
- It involves **experimenting with diverse algorithms**, fine-tuning parameters, and evaluating performance to achieve optimal results



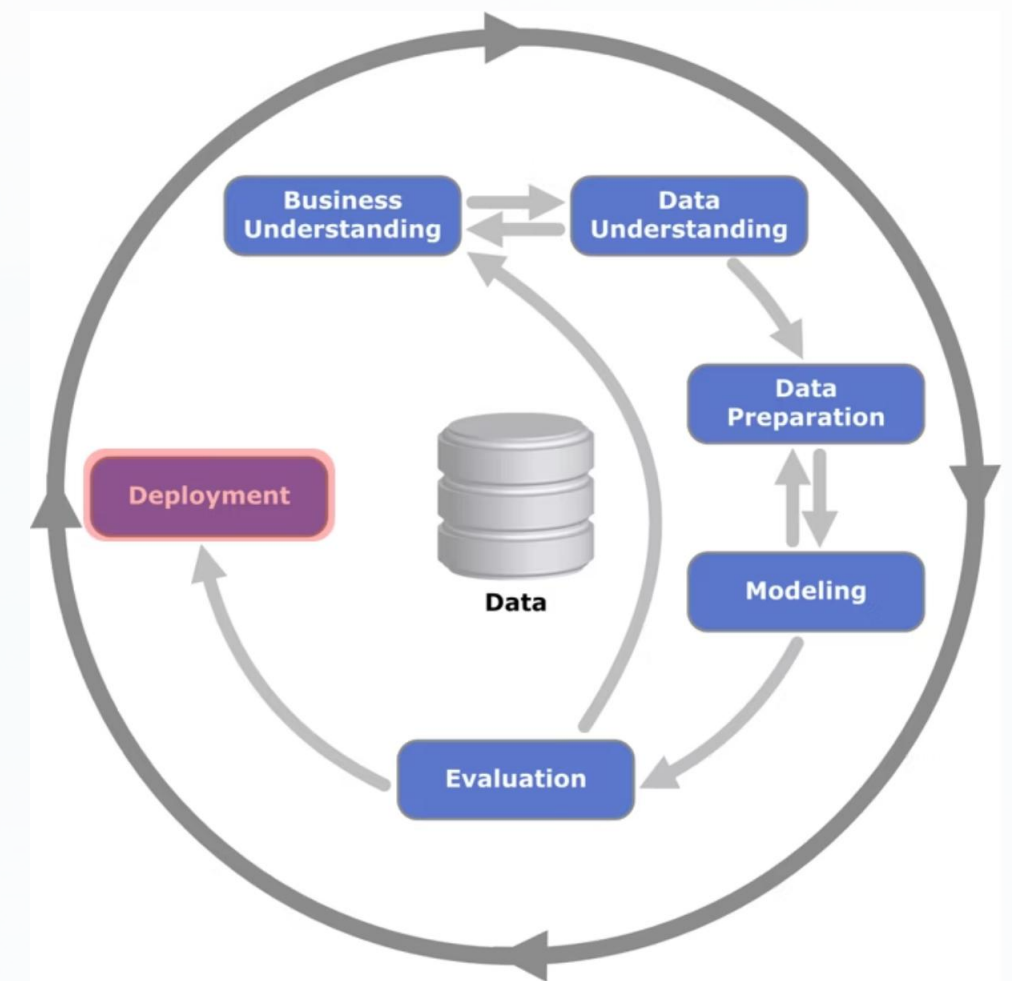
# The CRISP-DM process

- The **evaluation phase** focuses on assessing the quality and validity of the selected model(s)
- While initial assessments may indicate a high-quality model, this phase delves deeper, **employing various evaluation methods** to determine the model's true performance and identify the best candidate among alternatives
- This involves analyzing the model's predictive accuracy, stability, and generalizability using methods such as hold-out or cross-validation and performance metrics
- Beyond technical evaluation, this phase also involves a critical review of the model's alignment with business objectives



# The CRISP-DM process

- The **deployment phase** translates the model's insights into actionable outcomes, marking the transition from analysis to practical application
- This phase can vary in complexity, ranging from generating a comprehensive report to **integrating the model into operational systems** for ongoing decision support
- Clear communication of the model's results and its integration into existing workflows are critical for successful deployment
- Deployment often involves close collaboration between the data analyst and the customer, ensuring a smooth **transfer of knowledge** and a clear understanding of how to utilize the model effectively
- This may include training, documentation, and ongoing support to empower the customer to leverage the model's capabilities



# The CRISP-DM process

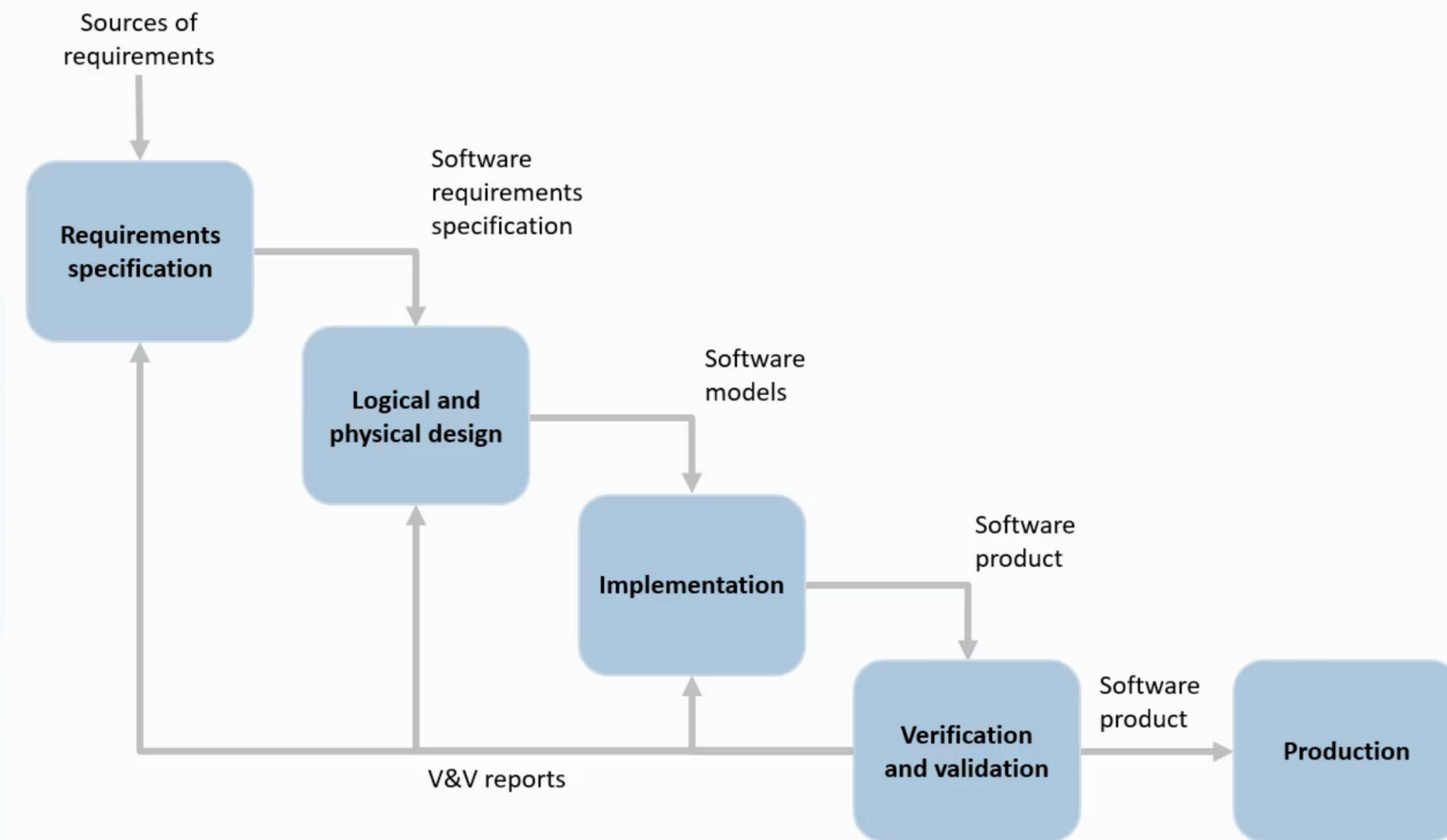
- Deployment establishes a **feedback loop**, where data collected from the deployed model is used to refine and improve future iterations, fostering a continuous cycle of learning and optimization
- A takeaway from CRISP-DM is recognizing that **data mining is an iterative journey**, not a linear race to a finish line
- Data scientists must embrace ongoing evaluation and refinement as new data emerges and algorithms evolve
- However, it is important to avoid getting trapped in the pursuit of perfection
- At some point, a decision must be made to deploy the current "best" model, accepting that incremental improvements may not outweigh the cost of delayed deployment





# Software development processes

## Classical Waterfall process



# Software development processes

## Agile Scrum process

